

DAY 20 TASK.py - C:/Users/krsb/OneDrive/Desktop/int

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```
#Task 1
```

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4, 5]
```

```
y = [10, 20, 30, 40, 50]
```

```
plt.plot(x, y)
```

```
plt.title("Simple Line Plot")
```

```
plt.xlabel("X-axis")
```

```
plt.ylabel("Y-axis")
```

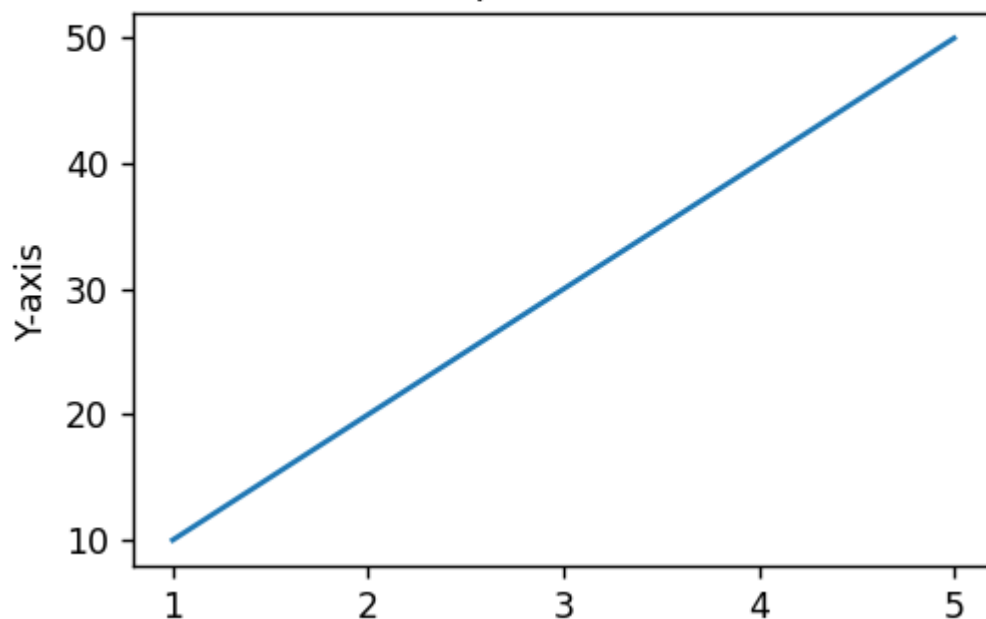
```
plt.show()
```



Figure 1



Simple Line Plot



#Task 2

```
import matplotlib.pyplot as plt
```

```
students = ["A", "B", "C", "D", "E"]
```

```
marks = [75, 85, 60, 90, 70]
```

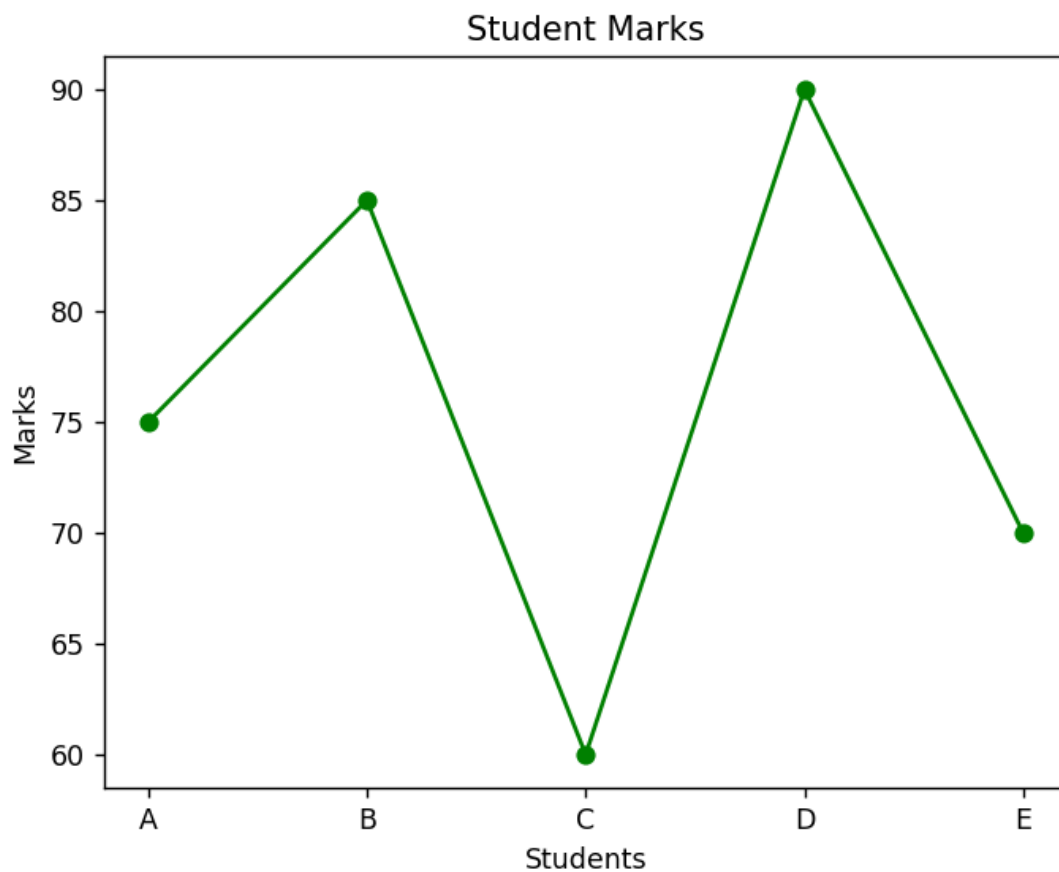
```
plt.plot(students, marks, marker="o", color="green")
```

```
plt.title("Student Marks")
```

```
plt.xlabel("Students")
```

```
plt.ylabel("Marks")
```

```
plt.show()
```

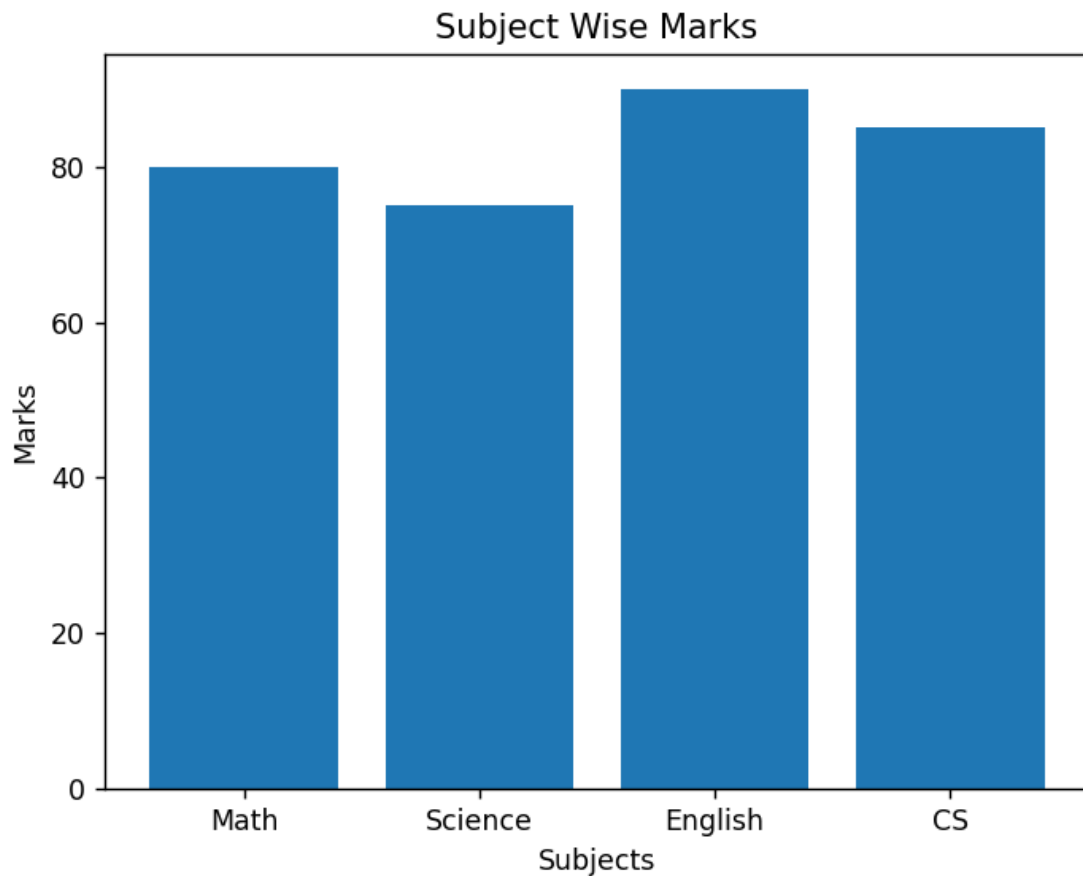


#Task 3

```
import matplotlib.pyplot as plt

subjects = ["Math", "Science", "English", "CS"]
marks = [80, 75, 90, 85]

plt.bar(subjects, marks)
plt.title("Subject Wise Marks")
plt.xlabel("Subjects")
plt.ylabel("Marks")
plt.show()
```

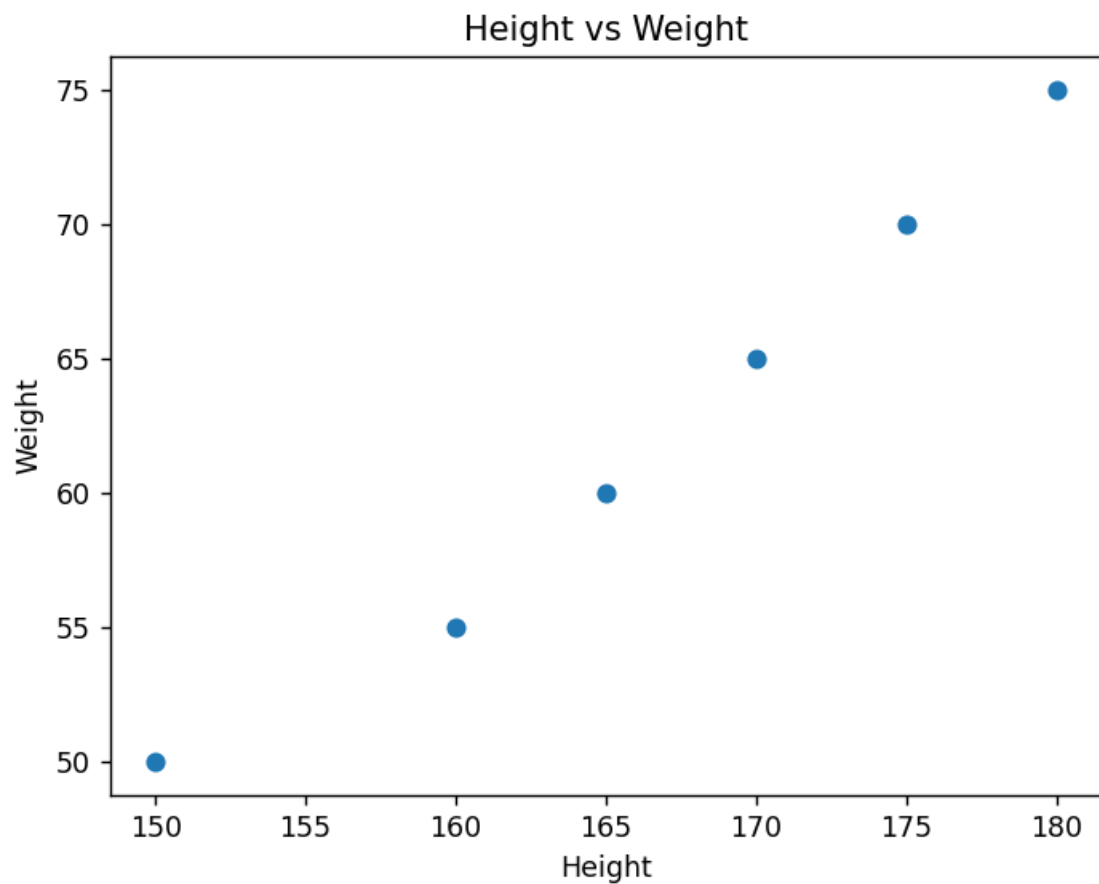


#Task 5

```
import matplotlib.pyplot as plt

height = [150, 160, 165, 170, 175, 180]
weight = [50, 55, 60, 65, 70, 75]

plt.scatter(height, weight)
plt.title("Height vs Weight")
plt.xlabel("Height")
plt.ylabel("Weight")
plt.show()
```

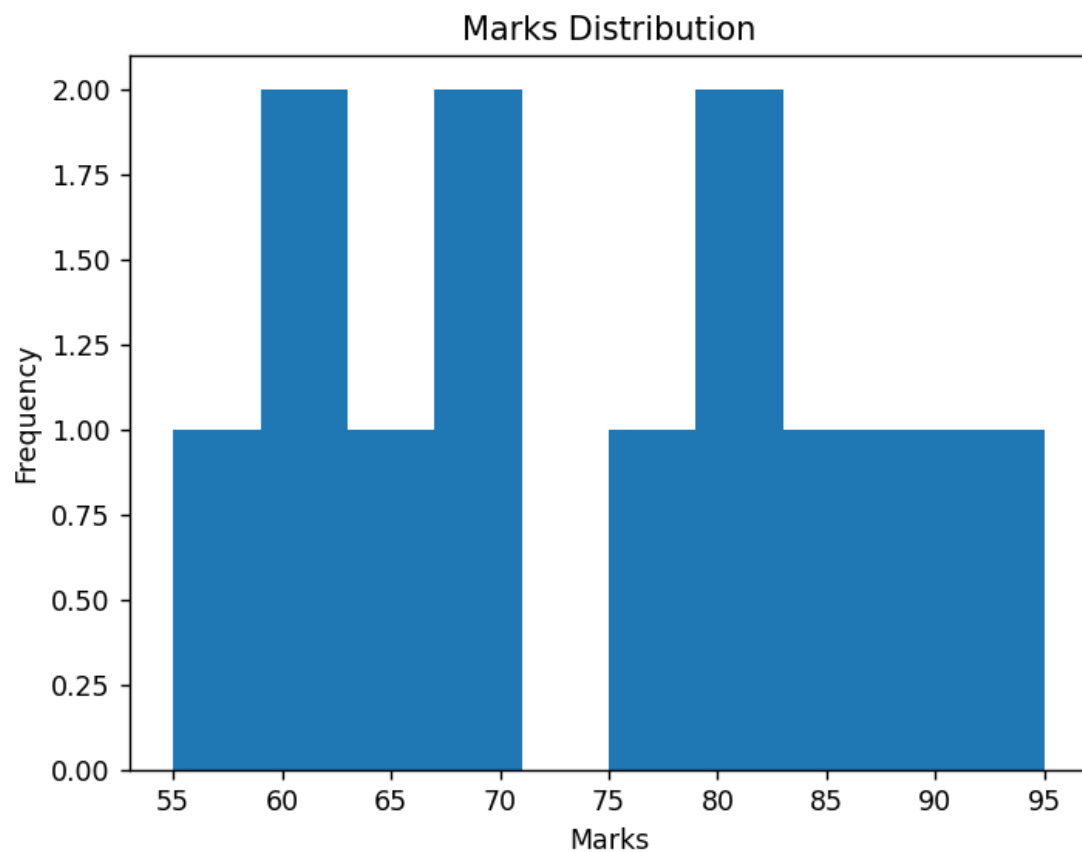


#Task 6

```
import matplotlib.pyplot as plt

marks = [55, 60, 65, 70, 75, 80, 85, 90, 95, 60, 70, 80]

plt.hist(marks)
plt.title("Marks Distribution")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.show()
```



#Task 7

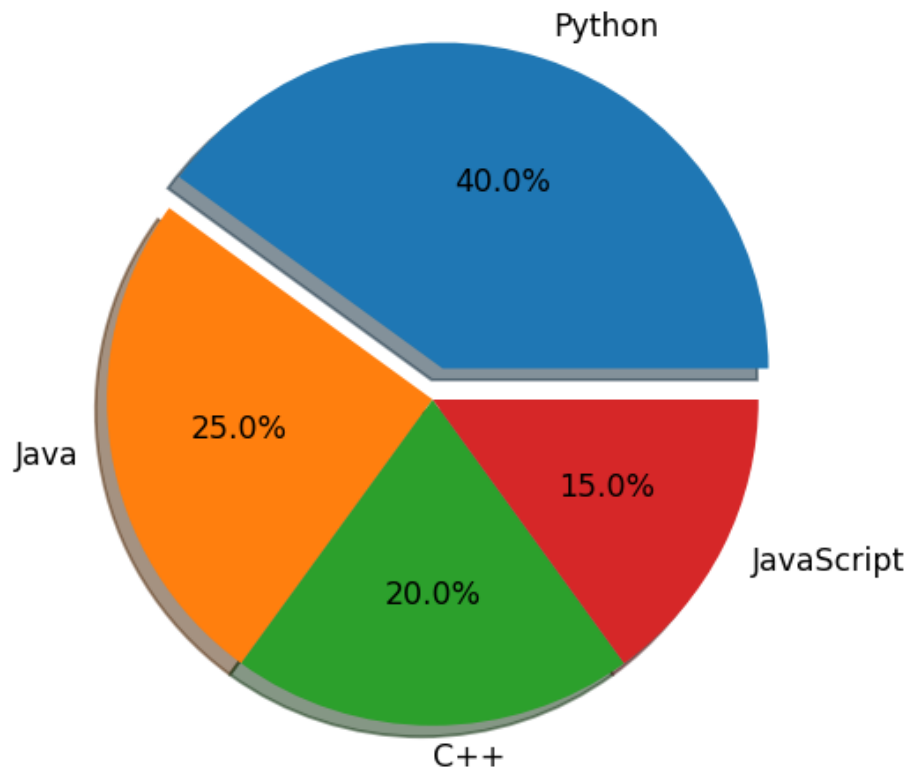
```
import matplotlib.pyplot as plt

languages = ["Python", "Java", "C++", "JavaScript"]
students = [40, 25, 20, 15]

explode = [0.1, 0, 0, 0]

plt.pie(students, labels=languages, explode=explode, shadow=True, autopct="%1.1f%%")
plt.title("Programming Language Popularity")
plt.show()
```

Programming Language Popularity

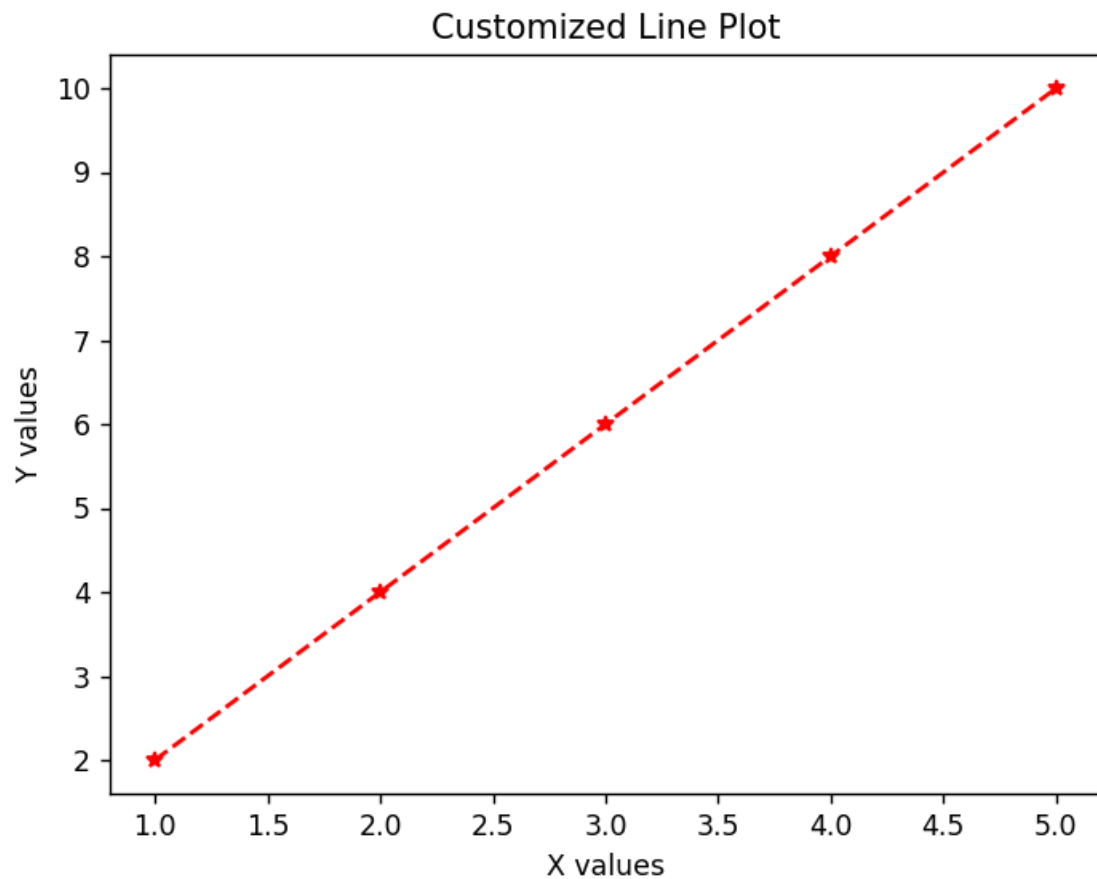


#Task 8

```
import matplotlib.pyplot as plt
import numpy as np

x = np.arange(1, 6)
y = x * 2

plt.plot(x, y, marker="*", linestyle="--", color="red")
plt.title("Customized Line Plot")
plt.xlabel("X values")
plt.ylabel("Y values")
plt.show()
```



#Task 9

```
import matplotlib.pyplot as plt
```

```
months = ["Jan", "Feb", "Mar", "Apr", "May"]  
sales = [200, 300, 250, 400, 350]
```

```
plt.plot(months, sales, marker="o")  
plt.title("Monthly Sales Data")  
plt.xlabel("Months")  
plt.ylabel("Sales")  
plt.show()
```

